

Final Assignment (English version)

1. The assignment must be submitted via e-mail to `l.bartolo@campus.lmu.de` with the subject line ‘Abschlussarbeit’.
2. The assignment must be submitted between **Monday, 21.09, 00:00** and **Thursday, 24.09, 23:59** (German local time). If you can only send the e-mail before that period, please contact me in advance.
3. The assignment is individual and must be written entirely either in German or entirely in English (see the separate file for the German version of the assignment).
4. The submission must include the *Declaration of Authorship* form (in the **Abschlussarbeit** folder), completed and signed, as required by the Faculty.
5. The submitted document must contain only your *Matrikelnummer*. Do **not** include your name anywhere in the document or in the file name.
6. The assignment must be submitted as a single file in **one** of the following formats:
 - a DOCX or ODT file containing typed text and math expressions (formulae, derivations, tables);
 - a ZIP file containing the source files (plus the PDF file) of a $\text{\LaTeX}$ document containing markup text and math expressions.

Mathematical expressions may be included in the files as scans of handwriting, but must be clearly legible. If you want to write the assignment fully in handwriting, please contact me sending a sample of your handwriting containing text and formulae in order to decide.

7. The sentences, formulae, or inferences you provide must not be identical or substantially similar to:
 - examples appearing in the class materials (including the exercise sheets);
 - previous solutions in this very assignment.

In particular, merely changing propositional variables or making superficial syntactic modifications while preserving the same logical structure does **not** count as a different example.

8. Regimentation (i.e., the use of parentheses) may depart from the exact formation rules introduced in the course to make a formula easier to read.

Exercises

Remark: The maximum number of points is **100**.

1. (24 points) Provide four inferences and, for each of them, determine whether it is valid in WK, SK, and L3 using truth tables. Make sure that:

1. each proposed inference contains exactly two distinct propositional variable types and at least three distinct logical connective types;
2. for each of these systems, there is at least one valid inference and at least one non-valid inference (among the four inferences you provide);
3. for every pair of systems, there is at least one inference that is valid in one system but non-valid in the other.

Provide, at the end, a table summarising which inferences are valid and non-valid in each system, following the style of Aufgabe 5.1.

2. (8 points) Provide two inferences that are valid in P0 and prove their validity by means of a Fitch derivation for P0 as introduced in the script. Make sure that:

1. each proposed inference contains at least three distinct propositional variable types and at least three distinct logical connective types;
2. each derivation uses at least three distinct derivation rules;
3. at least eight distinct derivation rules are used across the two derivations.

3. (8 points) Provide two inferences that are valid in M0 and then prove their validity by means of a Fitch derivation for M0 as introduced in the script. Make sure that:

1. each inference contains at least three distinct propositional variable types and at least three distinct logical connective types;
2. each derivation uses at least one of M0's characteristic rules (i.e., $\neg^+$ and $\perp^+$);
3. each of these characteristic rules is used at least once across the two derivations.

4. (8 points) Provide two inferences that are valid in I0 and then prove their validity by means of a Fitch derivation for I0 as introduced in the script. Make sure that:

1. each inference contains at least three distinct propositional variable types and at least three distinct logical connective types;
2. each derivation uses I0's characteristic rule (i.e., $\perp^-$).

5. (8 points) Provide two inferences that are valid in $\mathbf{C0}_1$ and then prove their validity by means of a Fitch derivation for $\mathbf{C0}_1$ as introduced in the script. Make sure that:

1. each inference contains at least two distinct propositional variable types and at least three distinct logical connective types
2. each derivation uses the rule $\perp^{1+}$.

6. (8 points) Provide two inferences that are valid in $\mathbf{W0}$ and then prove their validity by means of a Fitch derivation for $\mathbf{W0}$ as introduced in the script. Make sure that:

1. each inference contains at least three distinct propositional variable types and at least three distinct logical connective types;
2. each derivation uses at least one of the rules $\rightarrow^-$ and $\rightarrow^+$;
3. both $\rightarrow^-$ and $\rightarrow^+$ are used at least one across the two derivations.

7. (8 points) For each kind of modality covered in section 10.2 (namely, alethic, epistemic, deontic, and temporal modality), write one natural-language sentence expressing that kind of modality. Then formalise each sentence in the appropriate modal language. Make sure that:

1. you explain the intended meaning of each non-logical symbol;
2. each formula obtained from your formalisation contains at least three distinct propositional variable types;
3. each formula contains at least two occurrences of modal logical symbols;
4. each formula contains at least two occurrences of other logical symbols;
5. no sentence contains logical symbols belonging to more than one kind of modality.

8. (4 points) Write two sentences containing a definite description and formalise them in a language containing the descriptor ' ι '. Make sure that:

1. you explain the intended meaning of each non-logical symbol;
2. the descriptor ' ι ' is used at least twice in at least one formula;
3. each formula contains at least two occurrences of other logical symbols.

9. (4 points) Write two natural language expressions expressing complex properties and represent them with lambda abstracts (see Task 10.7 of the English script). Make sure that:

1. you explain the intended meaning of each non-logical symbol;
2. each formula contains at least two different predicate symbols;
3. each formula contains at least one predicate of arity greater than one;
4. each formula contains at least two additional logical symbols.

10. (20 points) The notion of a *sublogic* is defined as follows:

$$S \text{ is a sublogic of } T \stackrel{\text{DF}}{=} \mathbf{Fo}_S \subseteq \mathbf{Fo}_T \ \& \ \mathbf{Cn}_S \subseteq \mathbf{Cn}_T.$$

Determine whether each of the following statements is true or false and provide a short informal justification for your answer (see the examples below).

1. **M0** is a sublogic of **P0**.
2. **I0** is a sublogic of **S0**.
3. **S0** is a sublogic of **I0**.
4. $\forall n \in \mathbb{N} (\mathbf{C0}_n \text{ is a sublogic of } \mathbf{S0})$.
5. $\forall n \in \mathbb{N} (\mathbf{S0} \text{ is a sublogic of } \mathbf{C0}_n)$.
6. $\forall n \in \mathbb{N} (\mathbf{C0}_n \text{ is a sublogic of } \mathbf{C0}_{n+1})$.
7. $\forall n \in \mathbb{N} (\mathbf{C0}_{n+1} \text{ is a sublogic of } \mathbf{C0}_n)$.
8. Every logic is a sublogic of itself.
9. **S0** is a sublogic of **S1**.
10. **I0** is a sublogic of **S1**.

Examples:

1. **P0** is a sublogic of **S0**.

Answer: True. The following two points, together, provide a justification:

- **P0**'s language is a subset of **S0**'s language because the latter is defined similarly as the former, plus an additional clause for formulae containing ' $\neg$ '.
- **P0**'s consequence relation is a subset of **S0**'s consequence relation because the latter's derivation system contains all of the rules that **P0**'s derivation system contains (plus some others with ' $\neg$ ').

2. **S0** is a sublogic of **P0**.

Answer: False. Any of the following two points suffices as justification:

- **S0**'s language is *not* a subset of **P0**'s language because the former contains formulae with ' $\neg$ ', like ' $\neg p$ ', which is not contained in the latter.
- **S0**'s consequence relation is *not* a subset of **P0**'s consequence relation for many reasons. For instance, some inferences valid in **S0** – like $\neg\neg p/p$ – feature ' $\neg$ ', while **P0** has no such inferences.